

Cross-Period CMS Open-Data Replication of a Pre-Existing Wave Confinement Theory Log-Periodic Collider Prediction

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Abstract

Wave Confinement Theory (WCT) proposed prior to the present CMS analysis that curvature-induced spectral organization could generate discrete-scale, log-periodic structure in collider observables. In particular, a WCT preprint published on April 23, 2026 proposed a log-periodic deformation of the semileptonic Wilson coefficient $C_9(q^2)$ of the form

$$\delta C_9(q^2) = A \cos \left[\omega \ln \left(\frac{q^2}{\Lambda^2} \right) + \phi \right].$$

The present work does not measure C_9 directly. Instead, it performs an independent, model-agnostic search for related log-periodic structure in the inclusive opposite-sign dimuon invariant-mass spectrum of CMS Run-2 open data.

A development-stage scan over 2–120 GeV initially selected the lower scan boundary, revealing a broad-background degeneracy rather than a resolved oscillation. The analysis was therefore tightened by excluding known resonance windows from both continuum fitting and spectral testing, requiring frequencies corresponding to approximately two or more cycles across the analyzed log-mass range, applying the official CMS 2016 muon certification, and introducing both residual-permutation and parametric Poisson background-refit nulls.

In the certified Run2016H discovery file, an interior candidate was found at

$$\omega_m = 7.0258258258,$$

with amplitude $A_r = 0.7543$, phase $\phi = -0.1890$, and $\Delta\chi^2 = 75.76$. The candidate and complete analysis configuration were then frozen before independent-file testing. A second Run2016H file reproduced the frozen frequency with $A_r = 0.9367$, $\phi = -0.3060$, and $\Delta\chi^2 = 118.91$. More importantly, a separately preregistered Run2016G cross-period test reproduced the same frozen frequency with $A_r = 0.9349$, $\phi = -0.1568$, and $\Delta\chi^2 = 115.89$. In both replication samples, zero of 1000 residual permutations and zero of 500 parametric Poisson background-refit trials exceeded the observed fixed-frequency statistic, so the corresponding reported Monte Carlo p -values are resolution floors rather than precise tail probabilities.

These results constitute positive cross-period empirical support for the pre-existing WCT prediction class of collider log-periodicity. They do not, by themselves, establish WCT as the unique explanation. The principal remaining alternatives include stable CMS trigger/acceptance/reconstruction structure, incomplete Standard Model or continuum modeling, and generic discrete-scale mechanisms already known in the literature. The decisive next stage is

therefore not further exploratory frequency searching, but frozen tests across background families, trigger-stable regions, independent channels/experiments, and a WCT-derived dimensionless mapping that predicts frequency and phase before new data are inspected.

1 Introduction

Discrete-scale invariance (DSI) is a well-established mathematical mechanism by which continuous scaling is replaced by invariance under preferred scale ratios, naturally producing periodic corrections in the logarithm of a scale variable [?]. Log-periodic behavior has consequently appeared in many theoretical and empirical contexts. In high-energy physics, prior examples include DSI-inspired descriptions of hadron mass spectra [?], log-periodic decorations of LHC transverse-momentum spectra within complex- q or Tsallis-type descriptions [?, ?], and approximately periodic invariant-mass resonance towers in clockwork/linear-dilaton models [?].

Wave Confinement Theory (WCT) makes a more specific claim: confinement and curvature-driven spectral organization can produce discrete-scale structure as an emergent property of confined wavefields. A WCT preprint published on April 23, 2026 applied this idea specifically to the $b \rightarrow s\ell^+\ell^-$ sector and proposed a log-periodic deformation of $C_9(q^2)$ [?]. Thus the general prediction of collider log-periodicity predates the CMS analysis reported here.

The present analysis asks a deliberately narrower empirical question:

Does an inclusive CMS opposite-sign dimuon mass spectrum contain a stable, resolved log-periodic residual that survives certification, explicit resonance exclusions, background-refit null tests, a frozen independent-file replication, and a frozen cross-period replication?

The answer obtained under the present pipeline is positive. However, the interpretation must distinguish three levels:

1. detection of reproducible log-periodic structure in the analyzed CMS observable;
2. consistency with a pre-existing WCT prediction class;
3. unique attribution of that structure to WCT rather than detector, Standard Model, statistical, or other DSI mechanisms.

The first two are supported by the current analysis. The third remains open.

2 Prior WCT prediction

2.1 Broader WCT structure

A central scalar curvature functional used in WCT is

$$\Theta[\psi] = -\frac{\nabla^2\psi}{\psi + \epsilon e^{-\alpha|\psi|^2}}, \quad (1)$$

with a positive energy functional of the schematic form

$$E[\psi] = \int (|\nabla\psi|^2 + |\Theta[\psi]|^2) \, d^n x. \quad (2)$$

The WCT program investigates whether nonlinear curvature feedback, finite-band spectral selection, winding structure, and confinement can organize physical observables into preferred spectral structures.

The present CMS work does *not* derive $\omega_m = 7.0258$ from Eqs. (??)–(??). That is an important open theoretical step. What was predicted prior to this CMS test is the *class* of log-periodic collider structure.

2.2 April 2026 collider prediction

The April 2026 WCT preprint proposed

$$\delta C_9(q^2) = A \cos \left[\omega \ln \left(\frac{q^2}{\Lambda^2} \right) + \phi \right], \quad (3)$$

motivated by curvature-induced spectral structure in a confined wavefield [?]. In conceptual form, the proposed chain was

curvature feedback $\longrightarrow$ spectral band selection $\longrightarrow$ discrete-scale structure $\longrightarrow$ log-periodic modulation. (4)

Equation (??) was written for the $b \rightarrow s \ell^+ \ell^-$ phenomenology associated with $B^0 \rightarrow K^{*0} \mu^+ \mu^-$. The inclusive CMS DoubleMuon spectrum analyzed below is therefore **not a direct measurement of C_9** and should not be described as such. The CMS analysis instead probes whether a related log-periodic residual occurs in a distinct collider dimuon observable.

2.3 Coordinate convention

The CMS analysis uses

$$x_m = \ln \left(\frac{m_{\mu\mu}}{m_0} \right), \quad (5)$$

whereas the WCT C_9 construction and the associated LHCb analysis use

$$x_{q^2} = \ln(q^2). \quad (6)$$

Since $q^2 = m_{\mu\mu}^2$,

$$\ln q^2 = 2 \ln m_{\mu\mu}, \quad (7)$$

and therefore frequencies expressed in the two conventions obey

$$\boxed{\omega_m = 2k_{q^2}, \quad k_{q^2} = \frac{\omega_m}{2}.} \quad (8)$$

The frozen CMS value $\omega_m = 7.0258258258$ corresponds to

$$k_{q^2} = 3.5129129129. \quad (9)$$

Raw numerical frequencies from different log-coordinate definitions must not be compared without this conversion.

3 Prior art and competing mechanisms

The existence of log-periodic behavior is not unique to WCT. The relevant question is whether a competing mechanism predicts the same observable, coordinate dependence, frequency/phase stability, and cross-domain behavior.

3.1 Generic discrete-scale invariance

DSI generically yields complex scaling exponents and log-periodic corrections to scaling [?]. Therefore observation of a log-periodic residual alone cannot establish WCT. Generic DSI is best regarded as a mathematical class containing multiple possible physical realizations.

3.2 Hadron-spectroscopy DSI

Tatischeff applied fractal/DSI methods to hadron masses and widths, reporting log-periodic corrections in hadron spectroscopy [?]. This is relevant high-energy prior art, but it is not a prediction of the present inclusive CMS dimuon continuum residual and does not use the WCT confinement mechanism.

3.3 Complex- q /Tsallis log-periodicity

Wilk and Włodarczyk reported log-periodic oscillatory decorations in high- p_T spectra and discussed descriptions based on a complex nonextensivity parameter or a log-periodically varying scale [?, ?]. This is an important collider-specific alternative to any broad claim that WCT uniquely introduced collider log-periodicity. However, the primary observable and physical interpretation differ from the present dimuon invariant-mass residual.

3.4 Clockwork/linear-dilaton resonance towers

Clockwork/linear-dilaton phenomenology predicts a tower of approximately periodic resonant peaks and motivates Fourier-space searches in dilepton and diphoton invariant-mass spectra [?]. This is perhaps the closest prior collider model in terms of a structured dilepton mass spectrum. Nevertheless, a resonance comb approximately periodic in mass is not the same functional hypothesis as

$$r(m) \propto \cos[\omega \ln(m/m_0) - \phi]. \quad (10)$$

The two can and should be compared quantitatively rather than conflated.

3.5 Standard Model and detector/systematic structure

For the present CMS result, the most serious competing explanation is not necessarily another beyond-Standard-Model theory. Stable structures can arise from

- trigger thresholds and changing trigger efficiencies;
- reconstruction and identification efficiencies;
- detector acceptance versus $m_{\mu\mu}$;
- residual resonance tails or heavy-flavor structure;
- continuum background misspecification;
- correlated residuals not represented by a Poisson null;
- run-dependent or run-stable calibration effects.

Because Run2016G and Run2016H use closely related CMS reconstruction and trigger infrastructure, cross-period replication is stronger than a same-file fluctuation test but does not eliminate CMS-specific systematic explanations.

4 CMS open data

4.1 Run2016H

The discovery and first frozen replication use the CMS DoubleMuon Run2016H Ultra Legacy NanoAOD dataset [?]:

`/DoubleMuon/Run2016H-UL2016_MiniAODv2_NanoAODv9-v1/NANOAOD.`

The CERN Open Data record reports 48,912,812 events in 28 files from run numbers 281613–284044 [?].

The two analyzed files are

$$127C2975-1B1C-A046-AABF-62B77E757A86.root, \quad (11)$$

$$183BFB78-7B5E-734F-BBF5-174A73020F89.root. \quad (12)$$

4.2 Run2016G

The cross-period replication uses the CMS DoubleMuon Run2016G Ultra Legacy NanoAOD dataset [?]:

`/DoubleMuon/Run2016G-UL2016_MiniAODv2_NanoAODv9-v2/NANOAOD.`

The CERN Open Data record reports 45,235,604 events in 29 files from run numbers 278820–280385 [?].

The preregistered first file used for the cross-period test is

`05DD095C-F6C3-9A4F-9FB3-348A5A6403D5.root.`

4.3 Muon certification

All primary certified analyses apply

`Cert_271036-284044_13TeV_Legacy2016_Collisions16_JSON_MuonPhys.txt,`

the official CMS list of luminosity sections valid for analyses requiring valid muons [?]. The certification record covers both Run2016G and Run2016H.

5 Event reconstruction and selection

NanoAOD files are read with `uproot` and `awkward`. Muon four-vectors are reconstructed from the standard NanoAOD quantities. The selection used throughout the frozen analysis is

$$p_T^\mu \geq 4 \text{ GeV}, \quad (13)$$

$$|\eta^\mu| \leq 2.4, \quad (14)$$

$$\text{ID} = \text{tight}. \quad (15)$$

Opposite-sign muon pairs are formed, and their invariant mass is

$$m_{\mu\mu}^2 = (E_{\mu^+} + E_{\mu^-})^2 - |\mathbf{p}_{\mu^+} + \mathbf{p}_{\mu^-}|^2. \quad (16)$$

The analysis interval is

$$2 \text{ GeV} \leq m_{\mu\mu} \leq 120 \text{ GeV}, \quad (17)$$

divided into 350 logarithmically spaced bins.

5.1 Explicit resonance masks

The following mass windows are excluded from both the continuum fit and the spectral analysis:

$$2.90 \leq m_{\mu\mu} \leq 3.30 \text{ GeV}, \quad (18)$$

$$3.55 \leq m_{\mu\mu} \leq 3.85 \text{ GeV}, \quad (19)$$

$$8.50 \leq m_{\mu\mu} \leq 11.50 \text{ GeV}, \quad (20)$$

$$80.0 \leq m_{\mu\mu} \leq 100.0 \text{ GeV}. \quad (21)$$

These remove the dominant J/ψ , $\psi(2S)$, bottomonium-family, and Z -region structures from the tested continuum.

This exclusion is essential. Omitting a resonance from a background fit while retaining it in the residual spectral scan would deliberately generate a large structured residual.

6 Continuum model

Let N_i denote the observed count in mass bin i with center m_i . The smooth background is estimated by fitting

$$y_i = \ln(N_i + 0.5) \quad (22)$$

as a Chebyshev polynomial in

$$x_i = \ln m_i. \quad (23)$$

The frozen polynomial degree is seven.

The fit is iteratively robustified. At each iteration,

$$r_i = \frac{N_i - B_i}{\sqrt{\max(B_i, 1)}} \quad (24)$$

is computed, and bins beyond the declared clipping threshold are excluded from subsequent continuum fitting. Explicit resonance masks remain excluded throughout.

Only bins with a model expectation above the declared minimum count are retained for spectral testing.

7 Log-periodic spectral test

The residual model is

$$r(x) = c + a \cos(\omega x) + b \sin(\omega x), \quad (25)$$

equivalently

$$r(x) = c + A_r \cos(\omega x - \phi), \quad (26)$$

where

$$A_r = \sqrt{a^2 + b^2}, \quad (27)$$

$$\phi = \text{atan2}(b, a). \quad (28)$$

The scan statistic is the improvement relative to an intercept-only residual model:

$$\Delta\chi^2(\omega) = \chi_0^2 - \chi_\omega^2. \quad (29)$$

The exploratory scan is

$$3.1 \leq \omega \leq 80 \quad (30)$$

with 1000 linearly spaced trial frequencies.

7.1 Spectral-resolution condition

For a log-coordinate span

$$\Delta x = x_{\max} - x_{\min}, \quad (31)$$

the number of cycles represented by a trial frequency is

$$N_{\text{cycles}} = \frac{\omega \Delta x}{2\pi}. \quad (32)$$

Across the approximately $\Delta x \simeq 4.08$ analyzed span, $\omega \simeq 3.1$ corresponds to approximately two cycles. Frequencies below this scale were excluded from the final exploratory search because sub-cycle behavior is strongly degenerate with broad continuum curvature.

8 Null distributions

Two Monte Carlo null procedures are used.

8.1 Residual-permutation null

The analyzed residuals are permuted across the fixed log-mass coordinates. For every permutation the complete frequency scan is repeated and its maximum $\Delta\chi^2$ is recorded. This controls the look-elsewhere effect over the declared ω -scan under the assumption that fitted residuals are exchangeable.

For a frozen-frequency replication, the statistic at the predeclared frequency is also evaluated under each permutation.

The Monte Carlo estimate uses the add-one correction

$$\hat{p} = \frac{N_{\geq} + 1}{N_{\text{MC}} + 1}. \quad (33)$$

8.2 Parametric Poisson background-refit null

The second null propagates Poisson counting noise and continuum refitting. For each pseudo-experiment:

1. counts are drawn as

$$N_i^* \sim \text{Poisson}(B_i); \quad (34)$$

2. the degree-seven robust continuum is refit to N_i^* ;
3. new Pearson residuals are computed;
4. the exploratory scan and/or frozen-frequency statistic is evaluated.

This is stronger than simply shuffling residual bins because the continuum is re-estimated in every pseudo-experiment. It still does *not* simulate trigger, reconstruction, calibration, acceptance, correlated detector effects, or an alternative Standard Model continuum family.

9 Analysis development and rejection of the initial boundary solution

The first 100,000-event development run used a broad 2–120 GeV fit without the final resonance exclusions and scanned $0.5 \leq \omega \leq 80$. It selected

$$\omega_{\text{best}} = 0.5, \quad (35)$$

exactly the lower scan boundary, with an unphysical residual amplitude of order 10^2 .

The full log span is

$$\ln(120/2) = \ln 60 \simeq 4.094, \quad (36)$$

so $\omega = 0.5$ represents only

$$N_{\text{cycles}} \simeq \frac{0.5(4.094)}{2\pi} \simeq 0.326 \quad (37)$$

cycles. The candidate was therefore interpreted as broad background misspecification rather than a resolved oscillatory signal.

This negative development-stage finding motivated explicit safeguards:

- resonance windows excluded from both fitting and testing;
- an exploratory lower bound near the two-cycle resolution scale;
- explicit boundary-frequency warnings;
- residual RMS and maximum-residual diagnostics;
- official CMS muon certification;
- background-refit parametric nulls.

The initial $\omega = 0.5$ solution is not used as evidence for WCT.

10 Certified Run2016H discovery

With the final masks and official muon certification applied to the first Run2016H file, 100,000 events were read and 50,951 survived the certification within that input segment. The analysis contained 21,930 dimuon masses in the declared mass interval and 287 analyzed bins.

The exploratory scan selected

$$\boxed{\omega_m = 7.0258258258} \quad (38)$$

with

$$A_r = 0.7542594, \quad (39)$$

$$\phi = -0.1889538, \quad (40)$$

$$\Delta\chi^2 = 75.7616. \quad (41)$$

The candidate spans

$$N_{\text{cycles}} = 4.5652, \quad (42)$$

and is not on a scan boundary.

The residual diagnostics were

$$\text{RMS}(r) = 1.2865, \quad (43)$$

$$\max |r| = 6.3986. \quad (44)$$

Thus the residuals are substantially better behaved than in the original development scan but are not perfectly unit-normal.

Zero of 1000 residual-permutation trials exceeded the observed global scan maximum, and zero of 500 Poisson background-refit trials exceeded it. Using Eq. (??), the reported Monte Carlo floors are

$$\hat{p}_{\text{perm}} = \frac{1}{1001} = 9.9900 \times 10^{-4}, \quad (45)$$

$$\hat{p}_{\text{refit}} = \frac{1}{501} = 1.9960 \times 10^{-3}. \quad (46)$$

These values mean zero exceedances at the available Monte Carlo resolution; they are not precision measurements of the true tail probability.

11 Preregistration of the CMS replication

Before inspecting a second CMS file, the following were frozen:

- $\omega_m = 7.0258258258$;
- mass range 2–120 GeV;
- 350 logarithmic bins;
- $p_T^\mu \geq 4$ GeV;
- $|\eta^\mu| \leq 2.4$;
- tight muon ID;
- degree-seven continuum;
- the four explicit mass exclusions;
- 100,000-event read cap;
- 1000 residual permutations;
- 500 parametric background-refit trials;
- random seed 20260827.

The exploratory best frequency in a replication file is secondary. The primary replication statistic is the amplitude and $\Delta\chi^2$ obtained at the frozen frequency in Eq. (??).

12 Frozen Run2016H independent-file replication

The second Run2016H ROOT file was not used to determine the frozen frequency. After applying the same configuration, 100,000 events were read, 90,868 survived certification, and 39,230 masses fell within the full mass interval.

The exploratory scan of this replication file independently selected

$$\omega_{\text{explore,H2}} = 6.563964, \quad (47)$$

but the preregistered primary result is the fit at $\omega_m = 7.0258258258$:

$$A_r = 0.9367121, \quad (48)$$

$$\phi = -0.3059911, \quad (49)$$

$$\Delta\chi^2 = 118.9148. \quad (50)$$

Again,

$$N_{\geq,\text{perm}} = 0/1000, \quad (51)$$

$$N_{\geq,\text{refit}} = 0/500. \quad (52)$$

The phase shift relative to discovery is

$$|\Delta\phi_{\text{H2-H1}}| = 0.11704 \text{ rad} \simeq 6.71^\circ. \quad (53)$$

The principal warning is that the absolute continuum residual diagnostics are worse:

$$\text{RMS}(r) = 1.8187, \quad (54)$$

$$\max |r| = 12.4685. \quad (55)$$

Consequently, this sample strongly rejects the implemented Poisson smooth-null model but simultaneously demonstrates that the smooth background is not a complete statistical description of the data.

13 Preregistered Run2016G cross-period replication

Before inspecting Run2016G data, the same fixed frequency and analysis configuration were preregistered for a cross-period test.

In the first Run2016G file, 100,000 events were read and all 100,000 were within accepted muon-certified luminosity sections for the processed segment. There were 39,633 masses in the declared 2–120 GeV interval.

The secondary exploratory scan selected

$$\omega_{\text{explore,G}} = 7.410711, \quad (56)$$

whereas the primary fixed-frequency result at $\omega_m = 7.0258258258$ was

$$A_r = 0.9348797, \quad (57)$$

$$\phi = -0.1567923, \quad (58)$$

$$\Delta\chi^2 = 115.8921. \quad (59)$$

For the frozen statistic,

$$N_{\geq, \text{perm}} = 0/1000, \quad (60)$$

$$N_{\geq, \text{refit}} = 0/500. \quad (61)$$

The frozen phase differs from the original discovery phase by only

$$|\Delta\phi_{\text{G-H1}}| = 0.03216 \text{ rad} \simeq 1.84^\circ. \quad (62)$$

The amplitude differs by approximately 24% from the discovery value and is almost identical to the independent Run2016H replication amplitude.

The Run2016G residual diagnostics are

$$\text{RMS}(r) = 1.7240, \quad (63)$$

$$\max |r| = 8.4725. \quad (64)$$

As in the second Run2016H file, these values argue against treating the fitted continuum as a complete physical null model.

14 Summary of results

Table 1: Certified discovery and frozen replication results. “H1” denotes the Run2016H discovery file, “H2” the independent Run2016H replication file, and “G” the preregistered Run2016G cross-period file.

Quantity	H1 discovery	H2 frozen replication	G frozen replication
Events read	100000	100000	100000
Events after JSON	50951	90868	100000
Masses in range	21930	39230	39633
Analysis bins	287	287	287
Frozen ω_m	7.025826	7.025826	7.025826
Amplitude A_r	0.754259	0.936712	0.934880
Phase ϕ [rad]	-0.188954	-0.305991	-0.156792
$\Delta\chi^2$	75.7616	118.9148	115.8921
Residual RMS	1.2865	1.8187	1.7240
Maximum $ r $	6.3986	12.4685	8.4725
Permutation exceedances	0/1000	0/1000	0/1000
Refit-bootstrap exceedances	0/500	0/500	0/500

The three frozen-frequency amplitudes have mean

$$\overline{A_r} \simeq 0.8753 \quad (65)$$

with a sample coefficient of variation of approximately 12%. No combined significance is assigned because the samples share analysis choices and, in particular, the CMS periods share detector/reconstruction infrastructure.

Table 2: Exploratory best frequencies. These are secondary in the two replication datasets because the primary test is fixed at the preregistered $\omega_m = 7.0258258258$.

Sample	Exploratory best ω	Best $\Delta\chi^2$	Best amplitude
Run2016H discovery	7.025826	75.7616	0.7543
Run2016H file 2	6.563964	128.5539	0.9719
Run2016G file 1	7.410711	144.1300	1.0316

15 Figures

The following figures summarize the certified Run2016H discovery, the independent Run2016H replication, and the preregistered Run2016G cross-period replication. For replication samples, the frequency scan is preferred over the automatically generated `best_log_periodic_fit.png`, because the latter displays the exploratory best-fit frequency rather than the preregistered frozen frequency.

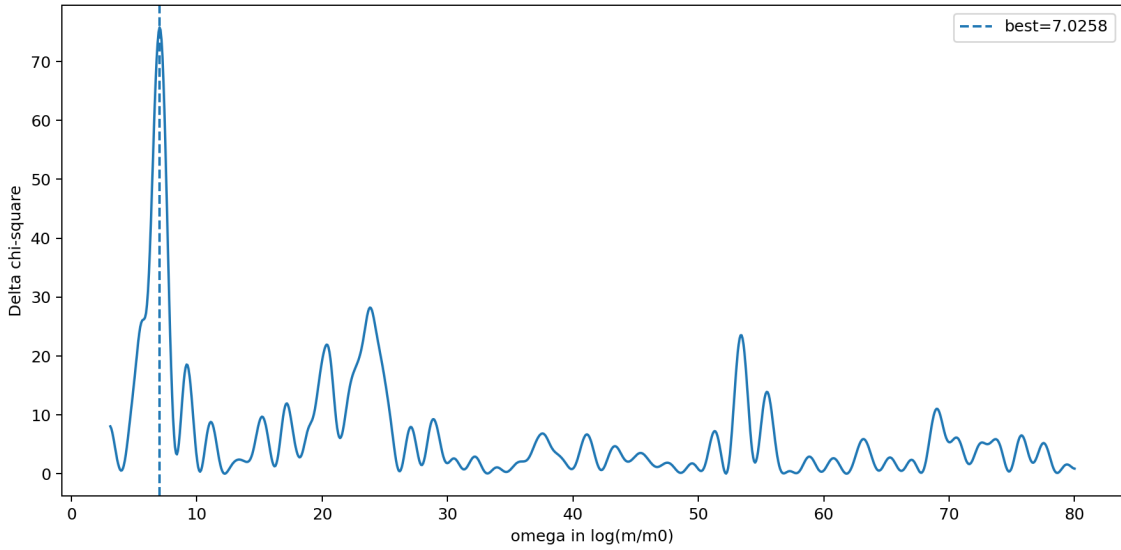


Figure 1: Certified Run2016H discovery frequency scan. The dominant resolved interior candidate occurs at $\omega_m = 7.0258258258$, with $\Delta\chi^2 = 75.76$ and an oscillation spanning approximately 4.57 cycles across the analyzed logarithmic mass interval. This value was subsequently frozen before the independent CMS replication tests.

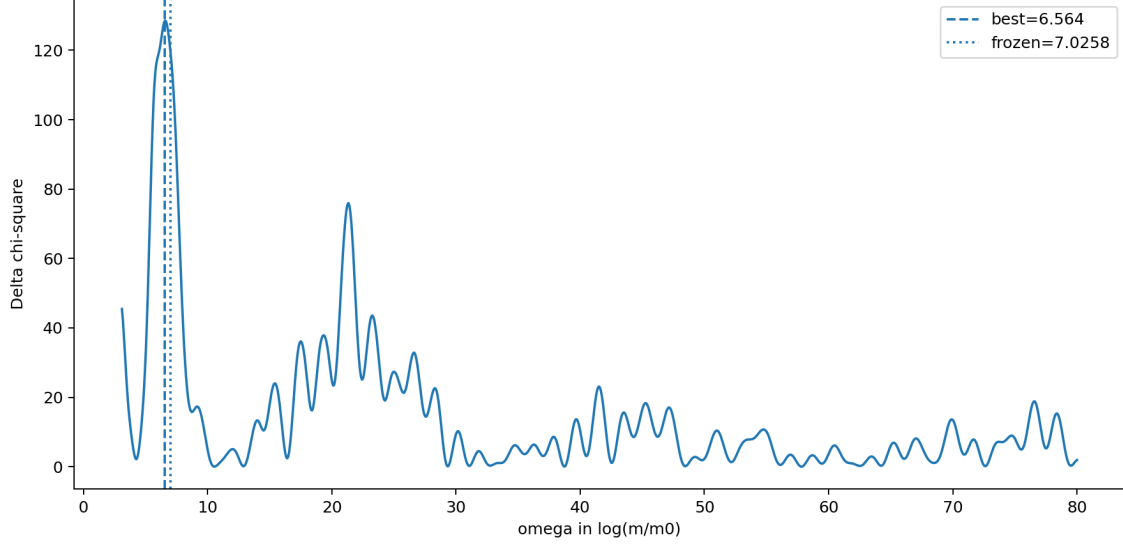


Figure 2: Independent Run2016H replication frequency scan. The dotted vertical line marks the preregistered frozen frequency $\omega_m = 7.0258258258$ obtained from the discovery sample. Although the secondary exploratory maximum occurs at $\omega = 6.5640$, the frozen frequency itself remains strongly supported, with $\Delta\chi^2 = 118.91$, amplitude $A_r = 0.937$, and zero exceedances in both 1000 residual-permutation trials and 500 parametric background-refit trials.

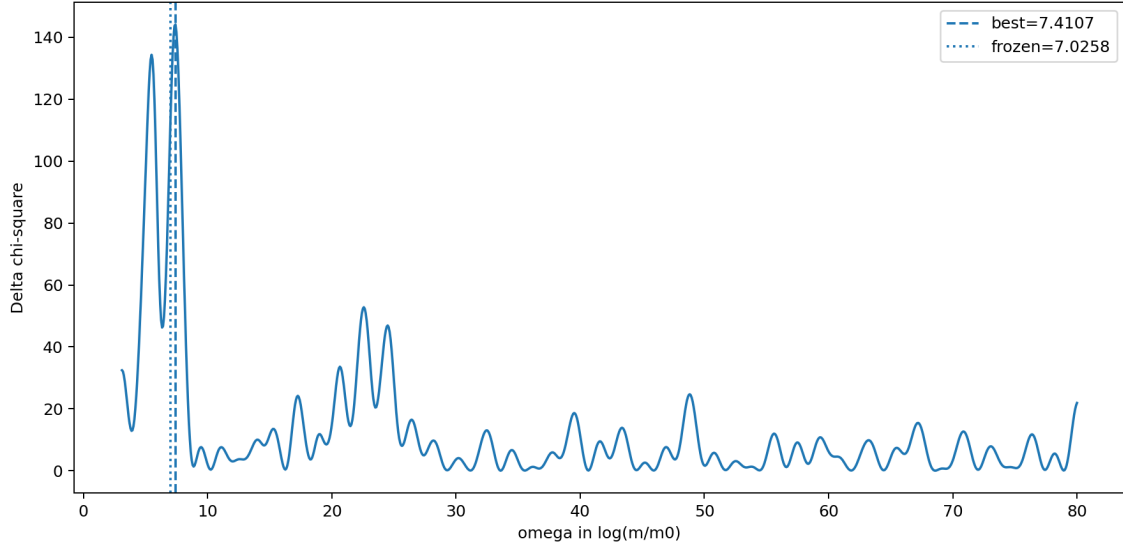


Figure 3: Preregistered Run2016G cross-period replication frequency scan. The dotted vertical line marks the frozen prediction $\omega_m = 7.0258258258$, fixed before inspection of the Run2016G sample. The independently selected exploratory maximum occurs at $\omega = 7.4107$, while the frozen frequency remains within the dominant spectral structure and gives $\Delta\chi^2 = 115.89$ with amplitude $A_r = 0.935$. Zero of 1000 residual permutations and zero of 500 parametric background-refit trials exceeded the observed frozen-frequency statistic.

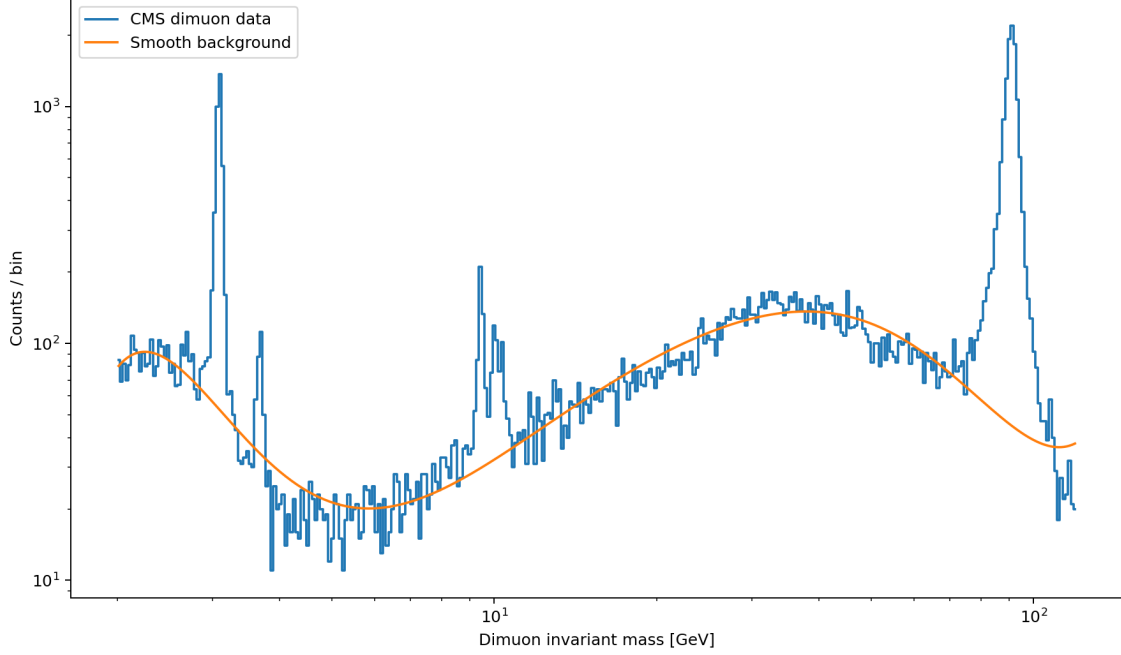


Figure 4: Certified Run2016G opposite-sign dimuon invariant-mass spectrum and the robust degree-seven smooth background model used to construct the Pearson residuals. The dominant J/ψ , $\psi(2S)$, bottomonium, and Z mass windows are excluded from the subsequent spectral analysis.

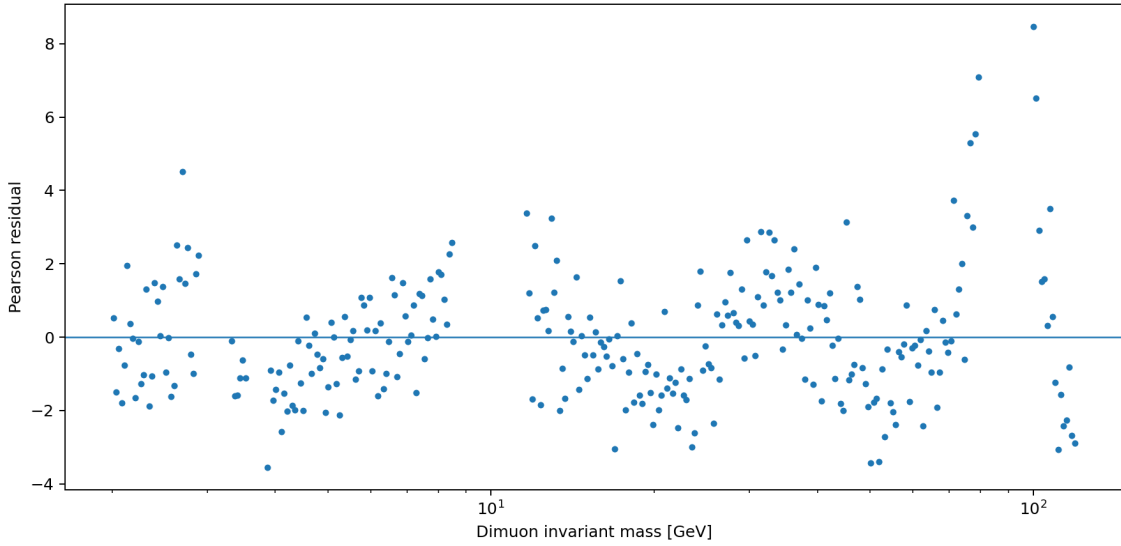


Figure 5: Pearson residuals of the certified Run2016G dimuon spectrum after the smooth-background fit and declared mass exclusions. The residual RMS is 1.724 and the maximum absolute residual is 8.473. The remaining structured variation demonstrates that the degree-seven continuum is not a complete physical null model and motivates the additional systematic controls discussed in the text.

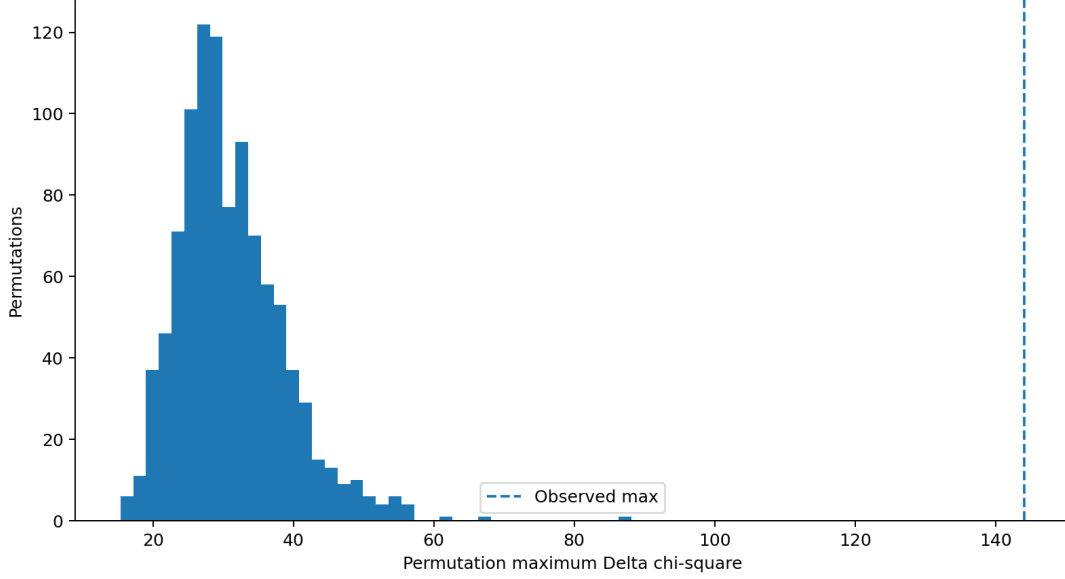


Figure 6: Residual-permutation distribution of the maximum exploratory $\Delta\chi^2$ obtained across the declared frequency scan in Run2016G. No permutation exceeded the observed scan maximum. This figure represents the global exploratory null distribution; the preregistered frozen-frequency significance is separately obtained from the fixed-frequency permutation and parametric-refit statistics.

16 Interpretation for WCT

16.1 What the data support

The CMS results support the following statement:

A log-periodic residual class proposed by WCT prior to the present CMS analysis is observed in a distinct inclusive CMS dimuon observable, and a CMS-discovered numerical candidate within that class subsequently reproduces at a preregistered fixed frequency in both an independent Run2016H file and a separately preregistered Run2016G cross-period sample.

This is stronger than a single exploratory pattern match for four reasons:

1. the broad low-frequency boundary artifact was explicitly rejected;
2. the surviving candidate is an interior spectral feature spanning more than four cycles;
3. the second Run2016H test used a frozen frequency;
4. the Run2016G test was frozen before that period was inspected.

The result is therefore properly described as

positive cross-period empirical support for the pre-existing WCT prediction class of collider log-periodicity.

16.2 What the data do not yet establish

The present result does not establish:

- that WCT is the unique origin of the CMS structure;
- that the CMS inclusive dimuon residual is a measurement of the $C_9(q^2)$ deformation in Eq. (??);
- that $\omega_m = 7.0258$ was derived from WCT field equations before the CMS data were analyzed;
- that the observed modulation survives a complete detector, trigger-efficiency, acceptance, and Standard Model treatment;
- that the same dimensionless WCT frequency has already been demonstrated across CMS, LHCb, gravitational-wave, photodiode, and atomic datasets.

These distinctions are not rhetorical qualifications; they define the next falsifiable tests.

17 Why the CMS result is not yet “case closed”

A repeated fixed-frequency signal can arise from either a reproducible physical effect or a reproducible systematic effect. Run2016G and Run2016H contain different collisions, and the cross-period replication therefore strongly weakens a one-off statistical-fluctuation explanation. However, both periods remain CMS 2016 data processed through closely related trigger, detector, and reconstruction systems.

The elevated residual RMS values,

$$1.29, \quad 1.82, \quad 1.72, \tag{66}$$

also show that the degree-seven smooth continuum does not fully whiten the spectrum. The refit-bootstrap result establishes that the observed fixed frequency is not reproduced by *Poisson fluctuations around that fitted continuum*. It does not establish that the fitted continuum is the correct complete null.

The strongest remaining mundane hypothesis is therefore

stable mass-dependent CMS/SM structure not represented by the current smooth null.

(67)

18 Specific systematic risks

18.1 Trigger and acceptance

The `DoubleMuon` dataset is selected by dimuon trigger paths and the current analysis does not yet enforce a single demonstrably flat trigger plateau over the entire 2–120 GeV interval or apply a mass-dependent trigger-efficiency correction. This is a major outstanding control. A stable efficiency modulation can reproduce across files and run periods.

18.2 Muon efficiency and reconstruction

Tight-ID, tracking, and reconstruction efficiencies can vary with muon kinematics. Since $m_{\mu\mu}$ is correlated with the individual muon p_T and η , efficiency structure can project into mass space.

18.3 Continuum model dependence

A single Chebyshev degree is not sufficient for a final physical inference. The result must survive a predeclared family of reasonable backgrounds without selecting the background that maximizes the WCT statistic.

18.4 Residual resonance and heavy-flavor structure

Explicit resonance windows remove the dominant peaks, but broad tails, interference, heavy-flavor components, and transitions between physical production regimes remain possible.

18.5 Correlated residuals

Neither independent residual permutations nor simple Poisson pseudo-counts model arbitrary bin-to-bin detector/systematic covariance. A final result requires structured nulls or simulation/control samples that encode such correlations.

19 Predeclared robustness matrix

A natural next test is a robustness matrix that is frozen before inspection. For example:

$$\text{background degree} \in \{5, 6, 7, 8, 9\}, \quad (68)$$

$$N_{\text{bins}} \in \{250, 300, 350, 400, 450\}, \quad (69)$$

$$\text{binning} \in \{\text{logarithmic, linear control}\}, \quad (70)$$

with the fixed frequency

$$\omega_m = 7.0258258258 \quad (71)$$

used as the primary statistic in every cell.

The purpose is not to demand that every implementation give identical significance. The relevant questions are:

1. Does the fitted amplitude retain a consistent sign and order of magnitude?
2. Does the phase remain stable?
3. Does the fixed-frequency statistic remain unusual under refitted nulls?
4. Is the result concentrated in a narrow analysis choice?

A family-wise null over the entire frozen robustness matrix would provide a much stronger statement than selecting the best-performing configuration.

20 Critical external replication tests

The most informative future tests are those that separate WCT from CMS-specific structure.

20.1 Different CMS channel

A dielectron, photon, or other independent observable could be transformed into a WCT-prescribed dimensionless coordinate and tested at a frequency frozen before inspection. Different trigger and reconstruction systems would substantially reduce shared-systematic explanations.

20.2 Independent experiment

An ATLAS or other independent collider dataset would be substantially more decisive because CMS-specific efficiency, reconstruction, and calibration features would no longer be shared.

20.3 Theory-derived frequency mapping

The most important theoretical milestone is to derive

$$\boxed{\text{WCT field parameters} \longrightarrow \text{dimensionless scale variable} \longrightarrow \omega, \phi, A} \quad (72)$$

before opening a new dataset.

This would convert WCT from predicting a broad phenomenological class (log-periodicity) to predicting a numerical cross-domain invariant. Such a test would distinguish WCT much more sharply from generic DSI.

21 Relation to the broader multi-domain WCT program

WCT has also been tested against gravitational-wave catalogs, laboratory photodiode data, atomic/NIST-related data, and LHC/LHCb observables. Those datasets should not be combined merely because each contains a spectral or log-periodic feature. A scientifically meaningful cross-domain claim requires:

1. a theory-defined dimensionless coordinate in every domain;
2. correct conversion between $\ln m$, $\ln q^2$, time, frequency, energy, or other coordinates;
3. frozen predictions before each new dataset is inspected;
4. explicit negative results and failed controls;
5. a joint null model that accounts for the number of datasets, observables, transforms, and analysis families searched.

If WCT derives a single invariant mapping and that mapping predicts the frequency and phase observed in multiple independently instrumented systems, the evidentiary status would be qualitatively stronger than independent post-hoc spectral matches.

22 Reproducibility

The analysis implementation is maintained in the `wct-cms` repository. The repository records:

- the discovery analysis;
- the frozen candidate document;
- the independent Run2016H replication result;
- the preregistered Run2016G replication;
- regression tests for Awkward-array selection;
- vectorized spectral scanning;

- residual-permutation nulls;
- parametric Poisson background-refit nulls.

At the time of the reported runs, the local regression suite returned

8 passed.

22.1 Discovery-file certified run

```
cms-wct \
--input data/files.txt \
--golden-json \
data/cert/14221/Cert_271036-284044
    _13TeV_Legacy2016_Collisions16_JSON_MuonPhys.txt \
--output-dir results/dimuon_certified_bootstrap500 \
--mass-min 2 \
--mass-max 120 \
--bins 350 \
--log-bins \
--muon-pt-min 4 \
--muon-eta-max 2.4 \
--tight-id \
--fit-degree 7 \
--mask-window 2.9:3.3 \
--mask-window 3.55:3.85 \
--mask-window 8.5:11.5 \
--mask-window 80:100 \
--omega-min 3.1 \
--omega-max 80 \
--omega-steps 1000 \
--max-events 100000 \
--permutations 1000 \
--parametric-bootstrap 500 \
--seed 20260827
```

22.2 Frozen Run2016H replication

```
cms-wct \
--input data/files_replication.txt \
--golden-json \
data/cert/14221/Cert_271036-284044
    _13TeV_Legacy2016_Collisions16_JSON_MuonPhys.txt \
--output-dir results/dimuon_replication_file2_frozen \
--mass-min 2 \
--mass-max 120 \
--bins 350 \
--log-bins \
--muon-pt-min 4 \
--muon-eta-max 2.4 \
--tight-id \
--fit-degree 7 \
--mask-window 2.9:3.3 \
```

```

--mask-window 3.55:3.85 \
--mask-window 8.5:11.5 \
--mask-window 80:100 \
--omega-min 3.1 \
--omega-max 80 \
--omega-steps 1000 \
--frozen-omega 7.025825825825827 \
--max-events 100000 \
--permutations 1000 \
--parametric-bootstrap 500 \
--seed 20260827

```

22.3 Frozen Run2016G replication

```

cms-wct \
--input data/files_run2016g.txt \
--golden-json \
data/cert/14221/Cert_271036-284044
_13TeV_Legacy2016_Collisions16_JSON_MuonPhys.txt \
--output-dir results/dimuon_run2016g_frozen \
--mass-min 2 \
--mass-max 120 \
--bins 350 \
--log-bins \
--muon-pt-min 4 \
--muon-eta-max 2.4 \
--tight-id \
--fit-degree 7 \
--mask-window 2.9:3.3 \
--mask-window 3.55:3.85 \
--mask-window 8.5:11.5 \
--mask-window 80:100 \
--omega-min 3.1 \
--omega-max 80 \
--omega-steps 1000 \
--frozen-omega 7.025825825825827 \
--max-events 100000 \
--permutations 1000 \
--parametric-bootstrap 500 \
--seed 20260827

```

23 Conclusion

The CMS analysis reported here began with a failed development-stage low-frequency boundary solution, which was rejected as a continuum-model degeneracy. After explicit resonance masking, a minimum spectral-resolution criterion, official muon certification, and stronger null procedures, a resolved Run2016H candidate emerged at

$$\omega_m = 7.0258258258.$$

That numerical candidate was frozen before an independent CMS file was inspected and reproduced at the fixed frequency in a second Run2016H file. The same frequency was then preregistered

before a Run2016G test and again produced a large fixed-frequency statistic, closely reproducing the original phase:

$$|\Delta\phi_{G-H1}| \simeq 1.84^\circ.$$

In both replication samples, zero of 1000 residual permutations and zero of 500 Poisson background-refit pseudo-experiments exceeded the observed fixed-frequency statistic.

Because WCT had publicly proposed collider log-periodicity before this CMS analysis, these results are **positive empirical support for a pre-existing WCT prediction class**. They are not yet a unique confirmation of WCT. Generic DSI, collider log-periodicity models, and especially stable CMS/Standard-Model mass-dependent structure remain viable alternatives. The elevated residual RMS in the replication samples makes that caveat quantitatively important.

The strongest next test is consequently not another unrestricted spectral search. It is a frozen discriminating program: trigger-stable selections, predeclared background-family robustness, independent detector channels, independent experiments, and a first-principles WCT mapping that predicts a dimensionless frequency and phase before new observations are opened.

References

- [1] R. J. Reyes, “A Curvature-Induced Log-Periodic Deformation of $C_9(q^2)$: Wave Confinement Theory and the LHCb $B^0 \rightarrow K^{*0}\mu^+\mu^-$ Anomaly,” Zenodo, Version 1, published April 23, 2026, DOI: [10.5281/zenodo.19705254](https://doi.org/10.5281/zenodo.19705254).
- [2] CMS Collaboration, “DoubleMuon primary dataset in NANOAOB format from RunH of 2016 (/DoubleMuon/Run2016H-UL2016_MiniAODv2_NanoAODv9-v1/NANOAOB),” CERN Open Data Portal, 2024, DOI: [10.7483/OPENDATA.CMS.UZD7.Z50M](https://doi.org/10.7483/OPENDATA.CMS.UZD7.Z50M).
- [3] CMS Collaboration, “DoubleMuon primary dataset in NANOAOB format from RunG of 2016 (/DoubleMuon/Run2016G-UL2016_MiniAODv2_NanoAODv9-v2/NANOAOB),” CERN Open Data Portal, 2024, DOI: [10.7483/OPENDATA.CMS.ZQS3.LGLP](https://doi.org/10.7483/OPENDATA.CMS.ZQS3.LGLP).
- [4] CMS Collaboration, “CMS list of validated runs Cert_271036-284044_13TeV_Legacy2016_Collisions16_JSON,” CERN Open Data Portal, 2023, <https://opendata.cern.ch/record/cms-14221>.
- [5] D. Sornette, “Discrete-scale invariance and complex dimensions,” *Physics Reports*, vol. 297, no. 5, pp. 239–270, 1998, DOI: [10.1016/S0370-1573\(97\)00076-8](https://doi.org/10.1016/S0370-1573(97)00076-8).
- [6] B. Tatischeff, “Fractal properties applied to hadron spectroscopy,” arXiv:1105.2034, 2011, <https://arxiv.org/abs/1105.2034>.
- [7] G. Wilk and Z. Włodarczyk, “Log-periodic oscillations of transverse momentum distributions,” arXiv:1403.3508, 2014, <https://arxiv.org/abs/1403.3508>.
- [8] G. Wilk and Z. Włodarczyk, “Tsallis Distribution Decorated With Log-Periodic Oscillation,” arXiv:1501.02608, 2015, <https://arxiv.org/abs/1501.02608>.
- [9] G. F. Giudice, Y. Kats, M. McCullough, R. Torre, and A. Urbano, “Clockwork / Linear Dilaton: Structure and Phenomenology,” *JHEP* 06 (2018) 009, arXiv:1711.08437, <https://arxiv.org/abs/1711.08437>.